

Query-Conditioned Evidence Allocation for Efficient RAG

Project Specification and Execution Plan v1.0 (QCCA)

CS787 Graduate Research Project • Benchmark: allenai/qasper • Hardware: 1x NVIDIA TITAN RTX (24GB)

Core Research Question: *Does retrieval uncertainty/structure predict how much uncompressed natural-language evidence a query requires, and can a lightweight query-conditioned allocator dynamically choose the amount of uncompressed context given to the generator while maintaining answer quality and reducing computational/context cost?*

1. Research Motivation

The broader research paradigm is **Query-Conditioned Resource Allocation**. While prior work addresses internal Transformer capacity (e.g. dynamic layer routing), this project addresses **information/evidence allocation** in Retrieval-Augmented Generation (RAG). Standard RAG presents all retrieved evidence in full natural language (high token and computational cost). SARA introduces context compression by fixing a split (α text passages, remaining $1-\alpha$ compressed as dense representations). QCCA asks whether α should dynamically depend on query uncertainty: $\alpha = \alpha(q)$.

2. Main Hypotheses & Falsification Conditions

H1 — Query-Dependent Evidence Requirement: Different queries require different amounts of uncompressed evidence; optimal $\alpha^*(q)$ is non-constant across queries.

H2 — Retrieval Structure Predicts Evidence Requirement: Features derived from the BM25 retrieval-score distribution (concentration, margin, entropy) contain predictive signal for required context.

H3 — Dynamic Allocation Improves Pareto Efficiency: A query-conditioned allocator achieves comparable or superior quality to the best static allocation while reducing context tokens ($\alpha > \alpha_{\text{Best Static } k}$).

Falsification Criteria: If optimal α does not vary across queries, if retrieval features have zero predictive signal, or if QCCA provides no Pareto advantage over $\{\alpha_{\text{dev}}\}^k$, the hypothesis is falsified and reported transparently.

3. Primary Benchmark Dataset

Benchmark: `allenai/qasper` (scientific research paper QA over long documents). Chosen because scientific papers require synthesis over extensive context. Single-dataset focus ensures high statistical power, rigor, and thorough failure mode analysis within the 8-week course timeline.

4. Paper-Disjoint Data Splitting & Leakage Control

Critical Protocol: Splits are partitioned strictly at the **paper level** (`example_id`), never at the individual question level. Questions from the same research paper must never cross development and test splits.

Partition	Papers	Questions	Designated Purpose
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Discovery	~ 11	~ 35	Exploratory hypothesis validation & sensitivity checks (locked; not used for final training).
Development	~ 90	~ 320	Fixed-k sweep, oracle generation, ϵ selection, GroupKFold allocator training.
Final Test	~ 180	~ 650	Untouched, frozen final evaluation arm. Zero parameter tuning after test begins.

5. Parent Architecture & Hardware Constraint

Parent architecture: **SARA** (BM25 retriever =10\$, Salesforce/SFR-Embedding-Mistral compressor =4096\$, mistralai/Mistral-7B-Instruct-v0.2 generator with XMistral projector).

24GB Titan RTX Constraint: Measured peak VRAM for SFR is \$ pprox 13.24 ext{ GB}\$; Mistral-7B is \$ pprox 13.56 ext{ GB}\$. Both cannot simultaneously reside in 24GB memory (6.8 ext{ GB} > 24 ext{ GB}\$). Execution is strictly **sequential**: passages are precomputed/cached via SFR, freed from VRAM, followed by Mistral generation. Latency is synthesized as { ext{total}} = L_{ ext{ret}} + L_{ ext{alloc}} + L_{ ext{comp}} + L_{ ext{gen}}\$ without disk I/O distortion.

6. Baseline Hierarchy

1. **Vanilla RAG (=10\$):** All passages as text, zero compression (quality ceiling).
2. **Pure Compression (=0\$):** All passages compressed into $\$ \angle ext{COMPRESS} \angle \$$ tokens (compression floor).
3. **Parent SARA (=5\$):** Fixed standard configuration.
4. **Best Static SARA ($=k_{ ext{dev}}^*$):** Optimal fixed $\in \{0,2,4,5,6,8,10\}$ selected on dev split only. (*Primary control*).
5. **Random Allocation Control:** \$ sampled uniformly from candidate set.
6. **RECOMP:** Positioned as conceptual related work (FLAN-UL2 20B is incompatible on 24GB VRAM and lacks QASPER support).

7. Fixed-k Sensitivity Sweep

All development questions are evaluated across $\in \{0, 2, 4, 5, 6, 8, 10\}$, logging Token F1, ROUGE-L, EM, input tokens, output tokens, isolated latencies, and peak VRAM to build the empirical training matrix.

8. Dual Oracle Formulations

- **Quality Oracle:** $^*_ { ext{qual}}(q) = \text{rg}\max_{k} ext{F1}(q, k)$ (ties broken to smaller \$). Represents diagnostic quality ceiling.
- **Efficiency-Aware Epsilon Oracle:** $^*_ {\epsilon}(q) = \min \{ k \mid ext{F1}(q, k) \geq \max_j ext{F1}(q, j) - \epsilon \}$ with $\epsilon \in \{0.00, 0.01, 0.02, 0.05\}$. Prevents context waste for trivial F1 gains. Frozen on dev set.

9. Metric Validation & Semantic Audit

Primary metric: Official QASPER Token F1 (normalized tokens). Secondary: ROUGE-L, EM. A blinded human review of 50 development questions at $\in \{2, 5, 10\}$ assesses correlation between Token F1 and human correctness (0/1/2 scale) before freezing oracle labels.

10. Retrieval Uncertainty Features

Calculated on rectified scores $s'_i = \max(s_i, 0)$:

- **Top-1 Concentration:** $ho_1 = s'_1 / (\sum_i s'_i + \epsilon)$
- **Top-2 Margin:** $\Delta_{12} = (s'_1 - s'_2) / (s'_1 + \epsilon)$
- **Retrieval Entropy:** $= \text{ext}\{\text{softmax}\}(s'_i / \text{au}), \{ \text{ext}\{\text{norm}\} \} = -\sum_i p_i \ln p_i / \ln(10)$
- **Query Length:** Word/token count in s .
- **High-Score Passage Count ($\{ \text{ext}\{\text{high}\} \}$):** Passages above relative threshold $\eta \cdot s'_1$.
- **Surface Category:** Boolean, Factoid, Explanatory heuristic (no gold data).

11. Edge-Case Robustness

Feature modules are unit-tested for: all-zero scores, equal scores, single dominant score, negative BM25 scores, and near-zero sums to guarantee numerical stability.

12. QCCA Allocator Policies

- **QCCA-Rule:** Interpretable piecewise policy (High confidence $\sigma_k=2$; Moderate $\sigma_k=4,6$; Diffuse $\sigma_k=8,10$). Thresholds tuned on dev set.
- **QCCA-Learned:** L2-regularized Logistic Regression & shallow Decision Tree ($\text{ext}\{\text{depth}\} \leq 4$). Evaluated via 5-fold `GroupKFold` grouped by paper ID.

13. Master 8-Week Timeline

Week	Milestone	Key Deliverables
W1	Infrastructure	Environment lock, SARA verification, split manifests, BM25 feature extractor, SFR caching.
W2	Dev Sweep	$\in \{0,2,4,5,6,8,10\}$ dev matrix evaluation, determine $\{ \text{ext}\{\text{dev}\} \}^*$, baseline results.
W3	Oracle & Audit	Quality and Epsilon Oracles, optimal σ distribution, 50-question human audit.
W4	Allocator Dev	Train QCCA-Rule & QCCA-Learned (GroupKFold), dry-run final test harness with dummy allocator.
W5	Freeze & Test	Lock all features, weights, σ^* , thresholds. Execute Final Test set evaluation.
W6	Statistics	Paper-clustered bootstrap ($=10\{, \}000$), Wilcoxon test, isolated & synthesized latency breakdown.
W7	Ablations	Feature ablations, Pareto frontier (F1 vs tokens), failure taxonomy, human evaluation.
W8	Manuscript	Final LaTeX tables, figures, paper draft, viva presentation prep.

14. Final Test System Matrix

System	Retriever	Compressor	Generator	k Allocation	Experimental Role
Vanilla RAG	BM25	None	Mistral-7B	$=10$	Full-text reference
Pure Compression	BM25	SFR	Mistral-7B	$=0$	Compression floor
Parent SARA	BM25	SFR	Mistral-7B	$=5$	Parent architecture
Best Static SARA	BM25	SFR	Mistral-7B	$=k_{\{ \text{ext}\{\text{dev}\} \}^*}$	Primary fixed baseline
Random Allocation	BM25	SFR	Mistral-7B	$\sim K$	Variance control

QCCA-Rule	BM25	SFR	Mistral-7B	(q)\$	Proposed rule policy
QCCA-Learned	BM25	SFR	Mistral-7B	(q)\$	Proposed ML policy
Quality Oracle	BM25	SFR	Mistral-7B	$\wedge^*_{\text{ext}\{\text{qual}\}}\text{\$}$	Diagnostic ceiling
Epsilon Oracle	BM25	SFR	Mistral-7B	$\wedge^*_{\backslash\epsilon}\text{\$}$	Efficiency ceiling

15. Statistical Analysis & Failure Taxonomy

Clustered Bootstrap: =10{,}000\$ iterations resampling research papers with replacement to compute paired difference $\Delta \text{ext}\{F1\} = \text{ext}\{F1\}_{\text{ext}\{QCCA\}} - \text{ext}\{F1\}_{\text{ext}\{\text{Baseline}\}}$ with 95% empirical confidence intervals.

Failure Classification: Deterministic categorization into (1) Over-compression, (2) Under-compression, (3) Retrieval failure (not in top-10), (4) Generator failure, and (5) Misleading retrieval distribution.

One-Line Summary: Instead of giving every RAG query the same amount of full-context evidence, QCCA learns whether the current query needs 2, 4, 6, or 8 passages in natural language while compressing the remaining evidence—and tests whether this adaptive allocation beats the best fixed allocation in the quality-vs-efficiency tradeoff.